## 1.2 Actors: A quick overview

Akka is centered on actors. Most of the components in Akka provide support in some way for using actors, whether for configuring actors, connecting actors to the network, scheduling actors, or building a cluster out of actors. What makes Akka unique is how effortlessly it provides support and additional tooling for building actor-based applications so you can focus on thinking and programming in actors.

Briefly, *actors* are a lot like message queues without the configuration and message-broker-installation overhead. They're like programmable message queues shrunk to micro size—you can easily create thousands or even millions of them. They don't "do" anything unless they're sent a message.

*Messages* are simple data structures that can't be changed after they've been created: they're *immutable*. Actors receive messages one at a time and execute some behavior whenever a message is received. Unlike queues, they can also send messages (to other actors).

Everything an actor does is executed asynchronously, meaning you can send a message to an actor without waiting for a response. Actors aren't like threads, but messages sent to them are pushed through on a thread at some point. How actors are connected to threads is configurable, as you'll see later; for now, just know that this is not a hardwired relationship.

We'll get a lot deeper into exactly what an actor is. At this point, the most important aspect of actors is that you build applications by sending and receiving messages. A message can be processed